

README - Simple Shell (Command Runner)

Project Overview

Simple Shell is a small system utility developed in C that demonstrates process creation and command execution in a Linux environment. The shell accepts user commands, creates child processes using `fork()`, executes commands using `execvp()`, and waits for completion before accepting new commands.

Features

- Execute Linux commands from a custom shell.
- Create child processes using `fork()`.
- Run commands using `execvp()`.
- Wait for process completion using `wait()`.
- Handle invalid commands gracefully.
- Exit the shell using the `exit` command.

Requirements

- Linux Operating System (Ubuntu recommended)
- GCC Compiler
- Standard C Libraries

Compilation

```
gcc shell.c -o shell
```

Run the Program

```
./shell
```

Sample Input

```
SimpleShell> pwd SimpleShell> ls SimpleShell> date SimpleShell> exit
```

Sample Output

```
SimpleShell> pwd /home/student SimpleShell> ls main.c README.md output.txt SimpleShell> date  
Tue Jun 3 12:30:45 IST 2026 SimpleShell> exit Exiting shell...
```

Error Handling

```
SimpleShell> abcxyz Command not found: abcxyz
```

Project Concepts

- Process Creation (`fork`)
- Program Execution (`execvp`)
- Parent-Child Synchronization (`wait`)
- Command Parsing
- Error Handling

Conclusion

This project demonstrates fundamental operating system concepts including process management and command execution. It serves as a beginner-friendly example of how Unix/Linux shells work internally.